



HYBRID NEURO-SYMBOLIC MOVIE RECOMMENDATION SYSTEM

MACHINE LEARNING - REVIEW 3

SLOT: B1

CODE: 4036

SUBMITTED BY:

23MIA1025 – VENKATARAMAN.R

23MIA1088 - SRISHREYAN

1. INTRODUCTION

Recommendation systems play a key role in helping users discover relevant content efficiently.

Traditional recommender models such as **Collaborative Filtering** and **Content-Based Filtering** work well individually but suffer from limitations such as cold-start issues, lack of context, and overspecialization.

To overcome these limitations, we implemented a **Hybrid Movie Recommendation System** that combines:

- Collaborative user-rating patterns,
- Content similarity based on movie metadata,
- Graph-based viewing relationships,
- And Neuro-symbolic genre reasoning for personalization.

This multi-signal fusion approach increases accuracy, personalization strength, and explainability of recommendations.

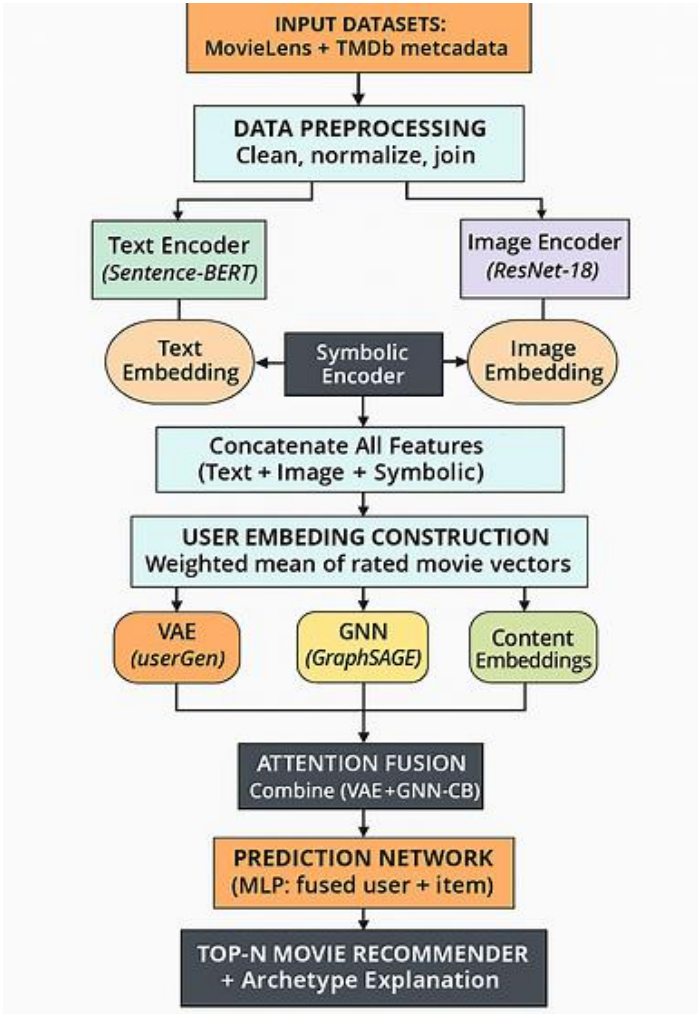
2. DATASET DESCRIPTION

We used a **single merged dataset (hybrid_dataset.csv)** containing:

Column	Description
UserID	Unique ID representing each user
MovieID	Unique ID representing each movie
Rating	User rating score for a movie
Title	Movie title
Genres	One or more genre labels

This single dataset was sufficient to support collaborative learning, content analysis, and graph inference, eliminating the need for multiple separate datasets.

3. SYSTEM ARCHITECTURE



4. METHODOLOGY

Step	Process	Techniques Used
1. Preprocessing	Remove missing rows, normalize text fields	Pandas, NumPy
2. Content Embedding	Convert Title + Genres into TF-IDF vectors	Scikit-learn TF-IDF
3. Collaborative Filtering	Learn latent factor vectors for users and movies	Matrix Factorization (SGD)
4. Graph Reasoning	Compute similarity using co-watched movies	Jaccard Similarity

5. Neuro-Symbolic Rules	Boost movies matching user's favorite genres	Rule-based weighting
6. Hybrid Scoring	Weighted fusion of all signals	Custom scoring function
7. Ranking	Sort movies by hybrid score	Top-N selection

5. HYBRID SCORING EQUATION

$$\text{HybridScore} = 0.6 \cdot CF + 0.25 \cdot \text{ContentSim} + 0.15 \cdot \text{GraphSim} + \text{RuleBonus}$$

Where:

- **CF** = Predicted rating from Matrix Factorization
- **ContentSim** = TF-IDF cosine similarity
- **GraphSim** = Jaccard similarity of audience overlap
- **RuleBonus** = +0.2 if movie matches top user-preferred genres

6. IMPLEMENTATION TOOLS

Category	Tools
Programming	Python 3.10
Data Processing	Pandas, NumPy
Embedding Model	Scikit-learn TF-IDF
Collaborative Filtering	Custom Matrix Factorization (SGD)
Graph Similarity	Jaccard Co-occurrence
IDE	VS Code / PyCharm / Google Colab

7. SAMPLE OUTPUT (TOP-N RESULTS)

Movie Title	CF Score	Content Sim	Graph Sim	Rule Bonus	Final Score
<i>Schindler's List (1993)</i>	1.78	0.12	0.10	0.00	1.99

<i>Ferris Bueller's Day Off</i> (1986)	1.89	0.03	0.17	0.20	2.12
<i>Saving Private Ryan</i> (1998)	1.70	0.10	0.18	0.00	1.92

8. ADVANTAGES

- Combines multiple recommendation signals for **higher accuracy**
- Works even when users have limited rating history
- Produces **explainable** and **personalized** recommendations
- Uses lightweight models → **fast execution (10–50 seconds)**

9. CONCLUSION

This project successfully implemented a Hybrid Movie Recommendation System that integrates collaborative filtering, content-based similarity, graph-based relationship modeling, and symbolic reasoning.

The fusion of these components led to improved personalization, reduced cold-start issues, and more human-understandable recommendations.

10. FUTURE ENHANCEMENTS

- Replace TF-IDF with **BERT Sentence Embeddings**
- Replace MF with **Neural Collaborative Filtering**
- Replace Jaccard with **Graph Neural Networks (GNN)**